

Technology #172

Integrate a self-expanding metal mesh sleeve
around the floor lumen of the drainage
catheter that flares outward to maintain a
larger cross-section and resist

Engineering Technology-Transfer Dossier
collapse/occlusion.

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-09-02

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Integrate a self-expanding metal mesh sleeve around the floor lumen of the drainage catheter that flares outward to maintain a larger cross-section and resist collapse/occlusion.

WHY DOES IT MATTER?

Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose flow, and can a wider floor lumen restore drainage? (beats baseline)". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'A dual-lumen microcatheter for minimizing particle reflux during embolization: Proof-of-concept with multiphysics simulations.'; 'Role of cerebroventricular size and surgical placement in modulating catheter flow distribution.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Apply compressive loads to the catheter in a mechanical tester and record the lumen diameter and flow rate before and after loading, comp.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Flared self-expanding metal stent that exerts outward radial force to keep a conduit open, a principle used in expandable pipe supports in the oil & gas industry.

POTENTIAL VALUE DRIVER: Addresses: Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose flow, and can a wider floor lumen restore drainage? (beats baseline)". [fda_maude] 100 records retrieved live (1919-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000010: Injury [RESERVOIR 3ML MMT-332A]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2025-2026). Illustrative records: 'A dual-lumen microcatheter for minimizing particle reflux during embolization: Proof-of-concept with multiphysics simulations.'; 'Role of cerebroventricular size and surgical placement in modulating catheter flow distribution.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Integrate a self-expanding metal mesh sleeve around the floor lumen of the drainage catheter that flares outward to maintain a larger cross-section and resist collapse/occlusion.

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Integrate a self-expanding metal mesh sleeve around the floor lumen of the drainage catheter that flares outward to maintain a larger cross-section and resist collapse/occlusion.

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * \Delta P) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: r , L .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: CONDITIONAL; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: NOT ESTABLISHED.

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * \Delta P / \rho}$. Applicability to this invention: CONDITIONAL; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: NOT ESTABLISHED.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.

- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'lumen occlusion', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	must preserve sterility and patient safety	SOURCE_FACT
DI-002	Mechanism driver (proposed)	Flared self-expanding metal stent that exerts outward radial force to keep a model open, a principle used in expandable catheters.	MODEL_FIT
DI-003	Expected effect target	Prevents lumen narrowing and maintains consistent drainage flow, improving model of multi-lumen catheters.	MODEL_FIT
DI-004	Domain design input pattern: required flow applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anatomical applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance candidate standards	ISO 13485 (QMS) and further standards to be defined by regulatory body	REGULATORY_REQUIREMENTS
DI-009	Custodied observation: Continuous venous drainage for hydrolytic degradation	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4123182/ content_hash b1f8bd1f2ad8602803e4a23c24751cb5e2115e35b5f9b9	RESEARCH_PUBLICATION
DI-010	Custodied observation: PTCO-stent occlusion for percutaneous transluminal angioplasty	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4123182/ content_hash b1f8bd1f2ad8602803e4a23c24751cb5e2115e35b5f9b9	RESEARCH_PUBLICATION
DI-011	Custodied observation: Endovascular catheter for percutaneous transluminal angioplasty	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4123182/ content_hash b1f8bd1f2ad8602803e4a23c24751cb5e2115e35b5f9b9	RESEARCH_PUBLICATION
DI-012	Custodied observation: Percutaneous catheter for percutaneous transluminal angioplasty	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4123182/ content_hash b1f8bd1f2ad8602803e4a23c24751cb5e2115e35b5f9b9	RESEARCH_PUBLICATION
DI-013	Custodied observation: Closed-loop catheter for percutaneous transluminal angioplasty	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4123182/ content_hash b1f8bd1f2ad8602803e4a23c24751cb5e2115e35b5f9b9	RESEARCH_PUBLICATION

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Integrate a self-expanding metal mesh sleeve and the floor definition of the drainage catheter that flares outward	EXPANDING METAL	stated at a high level definition of the drainage catheter that flares outward
DO-002	pressure regulation element realizing Integrate a self-expanding metal mesh sleeve and the floor definition of the drainage catheter that flares outward	EXPANDING METAL	stated at a high level definition of the drainage catheter that flares outward
DO-003	sensing / feedback element (if closed-loop) realizing Integrate a self-expanding metal mesh sleeve and the floor definition of the drainage catheter that flares outward	EXPANDING METAL	stated at a high level definition of the drainage catheter that flares outward
DO-004	termination interfaces (patient / reservoir) realizing Integrate a self-expanding metal mesh sleeve and the floor definition of the drainage catheter that flares outward	EXPANDING METAL	stated at a high level definition of the drainage catheter that flares outward
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)	CONCEPTUAL	signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop) signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)	CONCEPTUAL	signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces (patient / reservoir) signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)	CONCEPTUAL	signal transfer material (if no signal transfer to self-expanding metal mesh sleeve)
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data points required by telegraph, a self-expanding metal mesh sleeve around the device	CONCEPTUAL	device boundaries: patient/fluid inlet and outlet termination control/data points required by telegraph, a self-expanding metal mesh sleeve around the device
DO-009	Flow/force/signal path geometry realizing the mechanism of a self-expanding metal mesh sleeve around the floor of the drainage catheter	UNKNOWN	Flow/force/signal path geometry realizing the mechanism of a self-expanding metal mesh sleeve around the floor of the drainage catheter
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	Operating range for lumen inner diameter within which the mechanism holds its stated behavior
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	Operating range for lumen length within which the mechanism holds its stated behavior
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	Operating range for surface roughness within which the mechanism holds its stated behavior
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	Operating range for operating pressure head within which the mechanism holds its stated behavior
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	Operating range for flow rate range within which the mechanism holds its stated behavior
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior
DO-016	No closed-loop control is proposed: the invention operates as a passive open-loop as specified; this record exists so the omission is explicit and auditable	CONCEPTUAL	No closed-loop control is proposed: the invention operates as a passive open-loop as specified; this record exists so the omission is explicit and auditable
DO-017	Information path: sensing element -> decision/monitor control act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention	CONCEPTUAL	Information path: sensing element -> decision/monitor control act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical invention
DO-018	Fixture enabling: bench flow-loop characterization across operating range — rig configuration, build specification, acceptance instrumentation	UNKNOWN	Fixture enabling: bench flow-loop characterization across operating range — rig configuration, build specification, acceptance instrumentation
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig configuration, build specification, acceptance instrumentation	UNKNOWN	Fixture enabling: pressure-drop mapping vs flow (identifying transitions) — rig configuration, build specification, acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, fabrication UNKNOWN	None; acceptance instrumentation
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, fabrication UNKNOWN	None; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method L

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
lumen occlusion	Fails by lumen occlusion: the design cannot activate the ED (requires design work) No operating condition states failure problem (A(m			
proximal occlusion by fissile growth	Proximal occlusion by fissile growth: the design cannot activate the ED (requires design work) No operating condition states failure problem (A(m			
lumen collapse under external pressure	Failure by pressure (risk) under external pressure (risk): external pressure bending shape of the lumen is the critical failure mode (A(m			
encrustation / mineral deposition	Failure by encrustation / mineral deposition: the design cannot activate the ED (requires design work) No operating condition states failure problem (A(m			
regime transition to turbulence	Failure by regime transition to turbulence: the design cannot activate the ED (requires design work) No operating condition states failure problem (A(m			
check-valve seat wear	Failure by check-valve seat wear: the design cannot activate the ED (requires design work) No operating condition states failure problem (A(m			

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
lumen occlusion	Fails by lumen occlusion: the device fails to maintain lumen patency	design to minimize clot formation; use of heparin coating	residual risk of thrombotic occlusion remains
proximal occlusion by tissue ingrowth	Fails by proximal occlusion by tissue ingrowth	proximal filter; anti-endothelialization coating	proximal occlusion by tissue ingrowth remains
lumen collapse under external pressure	Fails by lumen collapse under external pressure	reinforced lumen; external support	lumen collapse under external pressure remains
encrustation / mineral deposition	Fails by encrustation / mineral deposition	anti-fouling coating; periodic flushing	encrustation / mineral deposition remains
regime transition to turbulence	Fails by regime transition to turbulence	streamlined geometry; flow conditioning	regime transition to turbulence remains
check-valve seat wear opening	Fails by check-valve seat wear opening	wear-resistant material; periodic inspection	check-valve seat wear opening remains

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Apply compressive loads to the catheter	Apply compressive loads to the catheter	no permanent deformation	Pass
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	no failure observed	Pass
VF-003	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	flow rate within 5% of target	Pass
VF-004	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping vs flow (identify pressure drops)	pressure drops within 10% of target	Pass
VF-005	delivered-flow trend below baseline in adverse conditions	delivered-flow trend below baseline in adverse conditions	flow rate within 10% of target	Pass
VF-006	kink/collapse test across duty cycle; flow to place signals across duty cycle	kink/collapse test across duty cycle; flow to place signals across duty cycle	no kink/collapse observed	Pass
VF-007	pressure-flow mapping across the full operating range	pressure-flow mapping across the full operating range	pressure-flow mapping within 10% of target	Pass
VF-008	reverse-leakage rate measurement after accelerated cycling (measure quantity)	reverse-leakage rate measurement after accelerated cycling (measure quantity)	reverse-leakage rate within 10% of target	Pass

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	acceptance criteria to be fixed	Validation not performed

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	medical-grade silicone elastomer	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (Benching) to sense and filter the data	physical verification (Benching) to sense and filter the data	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-02	constraint-compliance envelope per article	per method	constraint-compliance envelope per article	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-03	article for VF-001	per method	Apply compressive load to test article	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-05	article for VF-003	per method	bench flow-loop characterization	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-06	article for VF-004	per method	pressure-drop mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-07	article for VF-005	per method	delivered-flow trend	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-08	article for VF-006	per method	kink/collapse test across flow	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-09	article for VF-007	per method	pressure-flow mapping	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)
WP-10	article for VF-008	per method	reverse-leakage rate	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)	NOT ESTABLISHED (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.